

# Ezenia Diaz-Lembert

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## EDUCATION

**Harvard University** | Cambridge, MA | GPA: 3.549

Sep 2020-Feb 2025

Bachelor of Science in Electrical Engineering

**Relevant Coursework:** “Circuits, Devices, and Transduction”, “Signals and Communication”, “Applied Electromagnetism”, “Design of VLSI Circuits and Systems”, “Advanced Mixed-Signal CMOS Circuits”, “Neurophysiology and Neural Interfaces”

## TECHNICAL PROJECTS AND SKILLS

**Programs:** SystemVerilog, Virtuoso, Allegro System Capture, Cadence (Layout), LTspice, MATLAB, Simulink, Python, C, Bash scripting, Vivado, GIS

### **Dynamic Equalization Algorithm**

Jan 2024-Dec 2024

- Designed a real-time adaptive filter for personalized music equalization as a senior capstone project.
- Used a Convolutional Neural Network and spectral feature extraction on input audio signals for dynamic equalization.

### **MIPS Multicycle Processor**

Jan 2023-May 2023

- Designed and implemented a datapath and control unit for a multicycle processor in SystemVerilog.
- Tested stimulation and created testbenches, and synthesized onto an FPGA.

## WORK EXPERIENCE

**AVANGRID** | Augusta, ME

July 2025-Present

*Energy Control System Designer - Power System Analyst*

- Designing and maintaining transmission network model for Central Maine Power’s energy management system.
- Conducting studies on grid impact via contingency analyses.

**BOSE Corporation** | Framingham, MA

Jan 2025-May 2025

*Global R&D Consumer Systems and Architecture Co-op*

- Designing Simulink models for FPGA simulation and synthesis in Linux environment.
- Designing biquad filter models to emulate headset fits for FPGA-in-the-loop applications in testing and automation.
- Integrating external high-speed converters onto FPGA design for real time testing.
- Validating Simulink model FPGA deployment with an oscilloscope and waveform generator.

**Eaton Corporation** | Moon Township, PA

Jun 2024-Aug 2024

*Electrical Engineering Intern*

- Developed C code in IAR Tool to improve Total Harmonic Distortion computation accuracy from 7th to 39th current harmonics for the PCD-Air Circuit Breaker team.

**BAE Systems** | Merrimack, NH

Sep 2023-Jan 2024

*Electrical Engineering Intern*

- Conducted analog circuit design and analysis, simulations, and PCB layout for the Mixed Signal Integrated Circuit department.

**Halo Braid** | Cambridge, MA

Aug 2023-Sep 2023

*Electrical Engineering Intern*

- Developed a motor control system using an Arduino for an automatic hair braiding system for black hair.
- Leveraged encoders and mechanical rotation device parameters to optimize braiding pace, improving efficiency by 50%.

## LEADERSHIP, COMMUNITY INVOLVEMENTS, AND MEMBERSHIPS

**The Harvard-Radcliffe Veritones** | Cambridge, MA

Nov 2021-Aug 2024

*Social Chair/Alumni Coordinator*

- Organized events such as a 35<sup>th</sup> anniversary concert for more than 75 alumni and members of the a cappella group. Won first place at the International Championship of Collegiate A Cappella 2023 Quarterfinals Competition.

**Harvard SEAS Society of Hispanic Professional Engineers** | Cambridge, MA

Sep 2020-Dec 2023

*Corporate Liaison Officer*

- Organized accessible events and group initiatives for the Harvard SHPE chapter, including Latinx underclassmen/upperclassmen STEM talks and Problem Set nights.